Muhammed İkbal Yılmaz

Applied Econometrics | Time Series | Machine Learning

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Education

B.Sc. in Econometrics 2017 – 2024
Ankara Hacı Bayram Veli University, Ankara

Coursework (AA): Linear Algebra, Data Analysis, Sampling, Financial Econometrics, Quantitative Economic Models, International Macroeconomics and Econometric Models, Decision and Game Theory, Financial Management I.
Coursework (BA): Applied Time Series Analysis, Dynamic Analysis, Applied Operational Research.

Research

TÜBİTAK 2209-A Undergraduate Research Project 2023 – 2024
Brand preference and brand loyalty. Principal investigator; scholarship holder, 2023/1 call.
Survey design and data collection ($N = 407$); normality checks, Spearman correlations, χ^2 tests of independence, ANOVA with Tukey's HSD.

Classification of Brand Perception Using Random Forest: Brand Preference, Brand Loyalty, and Brand Trust
International Data Science and Statistics Congress (IDSSC), October 2024. Oral presentation in English, with Ö. Yıldız and Ş. Gökmen; published in the proceedings book.

Marka Tercihini Etkileyen Faktörler: Ankara Hacı Bayram Veli Üniversitesi Ekonometri ve İktisat Bölümü Lisans Programı Öğrencileri Üzerine Bir Uygulama
V. International Applied Statistics Congress (UYIK), 2024. Oral presentation in Turkish, with Ö. Yıldız and Ş. Gökmen; published in the proceedings book.

Selected projects

- **Return dynamics of ASELSAN.** Same-day relation of a stock's daily return with the market, the exchange rate, global risk and sovereign risk (ARX, Newey–West), and one-day-ahead forecasts compared across AR, ARX, Ridge, Lasso, random forest and XGBoost.
- **When is OLS BLUE?** Gauss–Markov assumptions checked one by one in a regression of monthly Bitcoin returns on macro-financial variables, and a random forest evaluated on months not used in fitting.

Other projects, with code, figures and project notes: github.com/mikbalyilmaz.

Skills

Software	Python (NumPy, pandas, statsmodels, SciPy, scikit-learn, XGBoost, Matplotlib), R, LaTeX
Languages	Turkish, English